

CS 4650/7650 ECE 4655/7655 Digital Image Processing Fall 2024

Homework 2: Segmentation using Otsu Thresholding [60 pts]

Out: Tuesday Sep 17

Due: Thursday Sep 26 (Midnight)

The goals of this assignment are:

- (1) implement Otsu binary thresholding;
- (2) understand dealing with color images;
- (3) understand limitations of global thresholding.

Programming Language: Python

Write a Python function to implement Otsu's binary thresholding method described in class. Specifically, implement "**Approach #3 Recursive Computation of Maximization of Between Class Variances**". This algorithm should automatically determine an intensity level to threshold an image to segment out the foreground pixels from the background. The output of your function should be a binary image that is black (pixel values = 0) for all background pixels and white (pixel values = 1) for all foreground pixels.

Apply this function to the given test images, report and discuss the results. Missing intermediate results (listed below) or missing discussion of the results will lead to deduction of points even if the submitted code is correct.

For each test image report should display:

- 1- Original image
- 2-Intensity histogram
- 3-Threshold value
- 4-Thresholded (binary) image
- 5-Brief discussion of the results.
 - a. Did the method generate a satisfactory result?
 - b. Why/Why not? If the method failed what is the main reason leading to failure?
- 6-Processing time

Note1: For color images, generate the above results for each color channel of the image. Specify color channel of choice and explain why.

Note2: For this homework you can use built-in histogram computation functions, but you can't use built-in Otsu thresholding functions (i.e. OpenCV cv.THRESH_OTSU, skimage.filters.try_all_threshold etc.).

Note3: Wikipedia (https://en.wikipedia.org/wiki/Otsu%27s_method) has a nice Otsu thresholding page with sample Matlab and Python codes. However, those codes correspond to Approach 1 (minimization of within-class variances) described in class. For this assignment you are supposed to implement Approach 3 described in lecture slides.

Report Structure: Your Written Report must conform to the following format.

1. Header
 - Course number and name
 - Assignment number and title
 - Your name
 - Date
2. Abstract (2 to 4 sentences describing the problem and the major results obtained)
3. Introduction (A brief statement of the experiments to be performed and the goal of the assignment)
4. Experiments and Results (figures, tables, graphs, ...). You need to use given test images.
5. Discussion/Conclusions
6. References/Appendix (As needed)

References

1. General tools that are useful to look at your input and output images: NIH ImageJ or Fiji which is an extension of ImageJ (opensource Java), imageMagick (opensource), gimp (opensource C++), Paint.Net, irfanview (shareware), Adobe Photoshop, Preview (MacOS), Windows PhotoViewer.
2. Canvas\Lect5_OtsuThresholding.pdf
3. <https://numpy.org/doc/stable/user/numpy-for-matlab-users.html>